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#include <stdio.h>

#define SIZE 19
#define WHITE 'O'
#define BLACK 'X'
#define EMPTY '_'

void inputGame(char (*game)[SIZE]);
void display(char (*game)[SIZE]);
char getWinner(char (*game)[SIZE]);

int main()
{
    char game[SIZE][SIZE];

    inputGame(game);
    //display(game);
    char answer = getWinner(game);
    switch(answer)
    {
        case BLACK:
            printf("Black");
            break;
        case WHITE:
            printf("White");
            break;
        default:
            printf("No winner");
            break;
    }

    return 0;
}

char checkHorizontal(char (*game)[SIZE])
{
    int count, i, j, k;

    char previous;
    for (i = 0; i < SIZE; i++)
    {
        previous = game[i][0];
        count = (previous == BLACK || previous == WHITE);
        for (j = 1; j < SIZE; j++)
        {
            if (game[i][j] == WHITE || game[i][j] == BLACK)
            {
                if (previous == game[i][j])
                    count++;
                else if (game[i][j] != EMPTY)
                    count = 1;
                else
                    count = 0;
            }

            if (count == 5)
                return previous;
        }
        previous = game[i][j];
    }
}

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    return EMPTY;
}

char checkVertical(char (*game)[SIZE])
{
    int count, i, j, k;

    char previous;
    for (i = 0; i < SIZE; i++)
    {
        previous = game[0][i];
        count = (previous == BLACK || previous == WHITE);
        for (j = 1; j < SIZE; j++)
        {
            if (game[j][i] == WHITE || game[j][i] == BLACK)
            {
                if (previous == game[j][i])
                    count++;
                else if (game[j][i] != EMPTY)
                    count = 1;
                else
                    count = 0;
            }

            if (count == 5)
                return previous;
        }
        previous = game[j][i];
    }

    return EMPTY;
}

char checkDiagonal(char (*game)[SIZE])
{
    int i = 0, j = 0, k, count;

    for (i = 0; i <= 14; i++)
    {
        for (j = 0; j <= 14; j++)
        {
            // Count diagonally if value is not empty
            if (game[i][j] != EMPTY)
            {
                count = 1;
                for (k = 1; k < 5; k++)
                {
                    if (game[i + k][j + k] == game[i][j])
                        count++;
                    else
                        break;
                }
            }
            else
                continue;

            if (count == 5)
                return game[i][j];
        }
    }
}

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    }

    for (i = 0; i <= 14; i++)
    {
        for (j = 4; j < SIZE; j++)
        {
            // Count diagonally if value is not empty
            if (game[i][j] != EMPTY)
            {
                count = 1;
                for (k = 1; k < 5; k++)
                {
                    if (game[i + k][j - k] == game[i][j])
                        count++;
                    else
                        break;
                }
            }
            else
                continue;

            if (count == 5)
                return game[i][j];
        }
    }

    return EMPTY;
}

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char getWinner(char (*game)[SIZE])
{
    char winner;

    // Check horizontally
    winner = checkVertical(game);
    if (winner != EMPTY)
        return winner;

    // Check vertically
    winner = checkHorizontal(game);
    if (winner != EMPTY)
        return winner;

    // Check diagonally
    winner = checkDiagonal(game);
    if (winner != EMPTY)
        return winner;

    return EMPTY;
}

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void inputGame(char (*game)[SIZE])
{
    int i = 0, j;
    char str[SIZE + 1];

    while(i < SIZE)
    {
        fgets(str, SIZE + 1, stdin);
    }
}

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        for (j = 0; j < SIZE; j++)
        {
            game[i][j] = str[j];
        }
        while (getchar() != '\n');
        i++;
    }
}

void display(char (*game)[SIZE])
{
    int i, j;

    printf("\n");
    for (i = 0; i < SIZE; i++)
    {
        for (j = 0; j < SIZE; j++)
        {
            printf("%c", game[i][j]);
        }
        printf("\n");
    }
}

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